

UNIVERSITY OF SOUTHERN CALIFORNIA

# Polygenic indices on All of Us

Session 2: from GWAS to PGI in the VM terminal

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30 minutes of instruction   30 minutes of practice

## What you will leave with

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- A polygenic index you computed with PLINK from external-style weights.
- A coverage report: which weights matched your data, and why some did not.
- An EDA table: distributions, correlation, and **incremental**  $R^2$ .
- Three figures: a histogram, kernel densities, and a decile gradient.
- A group comparison you can defend: balance where it should be, a shift where selection operates.

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|           |                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------|
| 0–9 min   | What a PGI is; where weights come from                                                            |
| 9–17 min  | Matching alleles, <code>-score</code> , standardization                                           |
| 17–28 min | Distributions, group comparisons, uses and limits                                                 |
| 28–60 min | Your own build: <code>score</code> , <code>explore</code> , <code>plot</code> , <code>save</code> |

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# Last session ended with 2,330 per-variant effects

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## Session 1

One regression per variant.

Each row: ID, counted allele, beta, SE, P.

One variant stood out; 2,329 did not.

## Today's move

Stop asking about one variant at a time.

**Aggregate** thousands of small effects into one number per person.

That number is the polygenic index.

For most social and behavioral traits, no single variant predicts anything useful; the *sum* of many tiny effects predicts meaningfully. That sum is the object social-science genomics actually uses.

**A PGI turns a GWAS you read into a variable you can analyze.**

# What a PGI is

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$$\text{PGI}_i = \sum_{j \in \text{matched variants}} w_j \times \underbrace{G_{ij}}_{0, 1, \text{ or } 2 \text{ copies of the effect allele}}$$

- $G_{ij}$ : the allele count you met last session.
- $w_j$ : a per-variant weight **estimated elsewhere**, in a discovery GWAS.
- The sum runs over the variants present in *both* the weight file and your data.

**Three ingredients: your genotypes, someone's weights, and an honest match between them.**

## Where the weights come from

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- From a **discovery GWAS** in a large external sample; for social-science traits, packaged releases like the **PGI Repository** provide ready-made weights per trait.
- **Never score the sample that produced the weights.** In-sample weights overfit: the PGI “predicts” noise it was trained on, and  $R^2$  flatters itself.
- Today’s `data/pgi_weights.txt` plays the external-consortium role: synthetic weights for  $\sim 2,000$  of our variants, shipped in a typical published format.
- Published weights are estimates, not truth: they carry the discovery sample’s noise, its ancestry mix, and its quality control.

**A PGI inherits every property of the GWAS behind it.**

## Matching alleles between files

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The weight file says: at SIMV000850, allele A, weight +0.031.

Your .pvar says: SIMV000850, REF A, ALT G.

- Which allele a weight refers to is a **fact stated in the file**, never an assumption. PLINK matches the *named* effect allele; orientation differences are handled for you.
- About 500 of today's weight rows name the other allele (sign flipped). The PGI is **identical** – same information, different bookkeeping.
- What is *not* recoverable: strand-ambiguous variants (A/T, C/G) with no stated strand, and **ID scheme mismatches** – 15 of today's rows use rsIDs our data has never heard of. They simply do not match.

**Name-based matching is robust; silent non-matching is the failure mode. Count your matches.**

# The anatomy of `-score`

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```
plink2 --pfile results/qc \  
  --score data/pgi_weights.txt 1 3 5 header \  
    cols=+scoresums list-variants \  
  --out results/pgi
```

|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| 1 3 5           | column numbers: variant ID, <b>effect allele</b> , weight |
| header          | first line is names, not a variant                        |
| cols=+scoresums | report the raw sum, not only the average                  |
| list-variants   | write exactly which variants entered the score            |

The log answers the coverage question: **1,922 processed, 79 skipped** – QC removals plus the 15 foreign rsIDs. Missing genotypes are **mean-imputed** by default: a documented choice, fine at our near-zero missingness.

# Standardizing and interpreting

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- Raw score units are meaningless. Convention: standardize to mean 0, SD 1 **within the analysis sample**, and speak in “SD of the PGI”.
- Association strength: correlation, or  $R^2$  from trait  $\sim$  PGI.
- The honest number is **incremental**  $R^2$ :  $R^2(\text{covariates} + \text{PGI}) - R^2(\text{covariates})$ . Today: 0.141 alone, but **0.063** incremental – less than half.
- A PGI is **sample-specific** after standardization, and **portability** is real: weights from one ancestry predict less well in another. AoU's diversity makes this a first-order concern, not a footnote.

**Report what the PGI adds, in a named sample, with its limits.**



## Look at distributions before you model

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- First questions for ANY constructed variable: how many people, what shape, where is the mass, are there outliers or clumps?
- For a PGI specifically: roughly bell-shaped (it is a big sum), no spikes (spikes suggest coverage or duplication problems), sensible tails.
- Then the same questions **within each group you plan to compare**.
- Today's tools: a summary table, histograms, kernel densities, and a decile plot – four views, one script, all written to `results/` as PNG files you open in the JupyterLab file browser.

**Every modeling surprise is cheaper to meet in a picture first.**

# Histograms and kernel densities

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## Histogram

Counts in bins. Faithful, granular, and the bin width is visible.

Best for: seeing actual data, spotting spikes and gaps.

## Kernel density

A small Gaussian bump on every point, summed. Smooth, comparable across groups.

Best for: overlaying groups. The **bandwidth is a choice** – too wide erases features, too narrow invents them.

```
bash scripts/08_pgi_eda.sh    # writes both, plus the decile figure
```

**Show a histogram to yourself, a density to your reader – and state the bandwidth choice.**

# Comparing groups honestly

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Today's data contain two group labels, built to disagree:

- treatment – **randomized**. PGI difference:  $-0.008$  SD. Balance is not a boring result; for an RCT it is a *finding* (and a design check).
- enrolled – **self-selected**. PGI difference:  $+0.46$  SD. Enrolling did not edit anyone's genome: selection correlates with population structure that leaked into the discovery weights.

The same picture – two densities – carries opposite meanings depending on **how the groups came to be**. The design question precedes the genetics.

**A PGI difference between groups is a fact about the groups, the weights, and the design – not automatically about biology.**

## What social scientists do with PGIs

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| Use                          | Discipline required                                         |
|------------------------------|-------------------------------------------------------------|
| Covariate / control          | Incremental $R^2$ ; measurement error means partial control |
| Gene–environment interaction | Both margins need causal identification; scale matters      |
| Balance check in RCTs        | Exactly today's treatment figure                            |
| Heterogeneous effects        | Prespecify; PGI quantiles are estimated, not observed       |
| Within-family designs        | Sibling differences remove structure and much confounding   |

**The PGI is a noisy, structure-carrying covariate – useful everywhere, causal nowhere by itself.**

## Small counts, same rules

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- Everything today is **aggregate**: summary tables, densities, decile means. That is the right default shape for outputs.
- The rules from session 1 still bind: no participant count of 1–20, including counts *derivable* from what you show. Group sizes in figure legends are counts too.
- Finer quantiles cut cells smaller: deciles of 588 people are fine; ventiles of a small subgroup may not be. The screen in `check_disclosure.py` runs at every save, and figures still need a **human** check.

**Aggregate by construction, screen by default, review by hand.**

## Your 30-minute lab

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- 30–35 Preflight; run session-1 QC if this is a fresh clone.
  - 35–42 Score the PGI. Read the log: how many matched, why not all?
  - 42–50 Run the EDA. Read the table; open the three figures.
  - 50–56 Answer the interpretation questions with a partner.
  - 56–60 Save the run; stop the app; exit check.
- 

```
bash scripts/00_preflight.sh      # then 02_qc.sh if results/qc is missing
bash scripts/07_score_pgi.sh
bash scripts/08_pgi_eda.sh
bash scripts/04_save.sh
```

## Questions to answer while your code runs

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1. Of 2,001 weight rows, how many scored? Name both reasons rows dropped.
2. Why is the PGI's incremental  $R^2$  so much smaller than its  $R^2$  alone?
3. The randomized arms differ by  $-0.008$  SD. Is that a null result or a finding?
4. The enrolled groups differ by  $+0.46$  SD. Give the honest causal reading.
5. Your decile figure is not monotonic everywhere. Problem, or expected?

**Extension: change the KDE bandwidth in `pgi_eda.py` and re-run. What changed, and which version would you publish?**

## The exit check

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“I can build a PGI, say what matched,  
read its distribution, and defend a group comparison.”

### **For your next project**

Fetch real weights for your trait (e.g., the PGI Repository) and state their discovery sample.

Decide the standardization sample and the covariates before looking at results.

Ask whether your comparison groups were assigned or self-selected – before interpreting any gap.